

Project report

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Entry No.: 2024MEB1337

Title:

Solid Modeling of a 6-piston radial engine in CAD

Objective:

To model and assemble a six-piston radial engine in CAD with accurate geometry and proper mechanical constraints.

Problem statement:

Radial engines use one crankshaft with a master rod and several smaller rods to move pistons placed in a circle. In CAD, you must set the parts and mates carefully so nothing collides and everything moves smoothly. The goal is to build a simple six-piston model, assemble it correctly, and check that the motion works.

Short description:

This project creates a 3D model of a 6-piston radial engine with all main parts, including the pistons, cylinders, crankshaft, crankcase, flywheel, and connecting rods. The parts were sized as needed for the model, assembled with the correct mates, and checked in a motion study to make sure the pistons move smoothly as the crank turns. The final output includes the completed assembly and detailed drawings of all components.

References:

1. https://en.wikipedia.org/wiki/Radial_engine
2. <https://users.metu.edu.tr/sonmez/MECH%20114/P%2056.pdf>